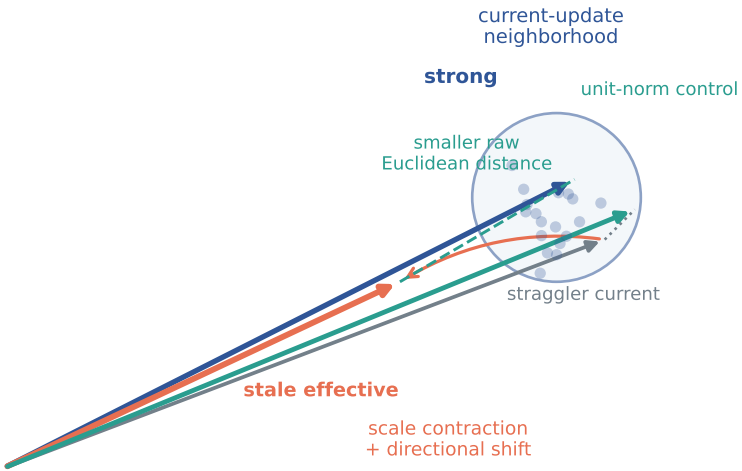
a Scale-sensitive selection geometry



Raw Euclidean distance mixes radius and angle;
normalized and cosine controls remove the radial channel.

b Crossed, auditable evidence chain

Frozen design

3 train × 3 system seeds
× 3 distance rules
NoAttack; logical staleness

Raw integrity

Resolved config + completion
300/60 sequential rounds
50 clients; 38 selected

Independent recomputation

Cell/window summaries
file-level SHA-256 hashes
no pseudo-replication

Early window

Euclidean attraction
scale-control attenuation

Converged boundary

system-seed-dependent decay
late invariance rejected

Directional boundary

stale cosine shift exists
stable exclusion rejected

Supported claim boundary

Scale-sensitive early selection attraction is reproducible
under the frozen simulation. Persistence and directional
interpretation depend on stage and resource assignment.

Not established: physical-device effects, attack robustness, fairness guarantees, or a new defense.